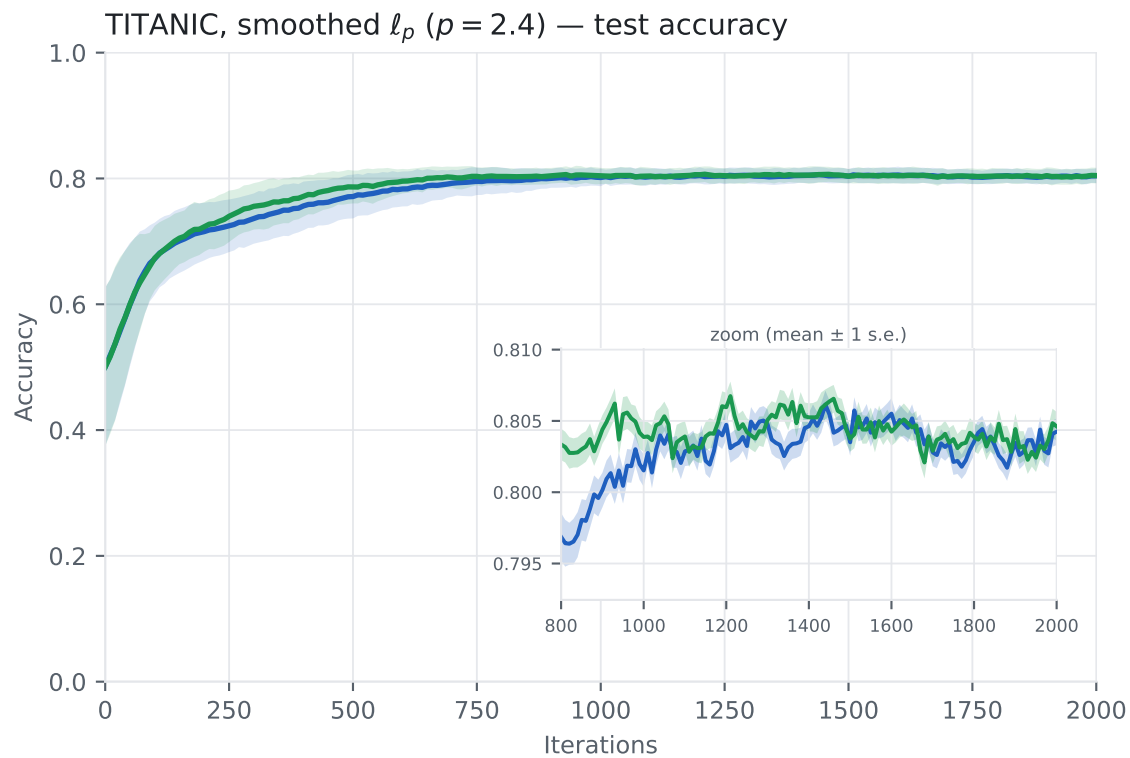
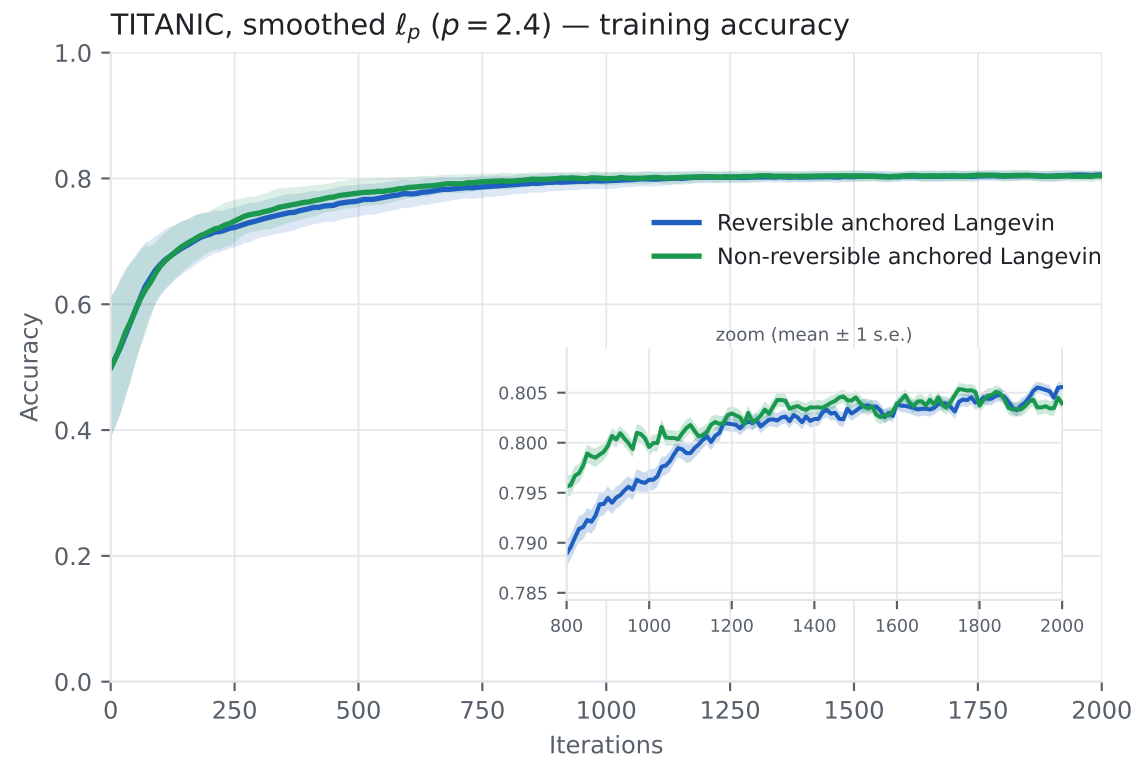


titanic_lp_eta3e-5: exact gradient, LASSO anchor



TITANIC -- smoothed ℓ_p , $g(w) = \sum_i (w_i^2 + \epsilon^2)^{p/2} \leq \text{Lambda}$, $p = 2.4$, $\epsilon = 0.18$, $\text{Lambda} = 4$. $d = 10$ (intercept w_0 + nine features), no mini-batching: the EXACT gradient is used at every step. $U = \sum \text{softplus} - y x \cdot w + w_0^2 / (2 \sigma^2) + \text{lambda} \sum_{j \geq 1} |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \delta^2}$; $a = \exp(U - U_0)$. $\sigma = 10$, $\text{lambda} = 7.12 = 0.01$ n_{train} , $\delta = 0.01082$, so $\bar{a}$ in $[0.50, 1]$. Block strengths $s = (5, 5, 5)$ on $w_1..w_9$ only. Step $\eta = 3e-05$. $R = 150$ coupled replicates (same W_0 , same Gaussian stream) on one fixed stratified 80/20 split ($n_{\text{train}} = 712$, $n_{\text{test}} = 179$). Paired final-iterate difference, non-reversible minus reversible: training -0.0016 ($t = -2.40$), test +0.0004 ($t = +0.36$). Bands: mean $\pm$ 1 SD (ddof = 1) across replicates -- repeat-run variability at a fixed split, not a confidence or credible interval. $\eta = 3e-5$ was SPECIFIED, not selected. $s = 5$ is shown; the full 4-strength sweep at this η is in the table above. Verdict at $s = 5$: no significant difference on the training metric.